

Smart Autonomous Shopping Cart Project – Logbook

Student Name: Mohammed Shalaby

Module: PDE4435 – Robotic System Integration

Role: Documentation & Report Development

Project Title: Smart Autonomous Shopping Cart Robot

24/3/2026

- Went through coursework requirements and discussed project direction with the team
- Started understanding the Smart Cart idea and its purpose

25/3/2026

- Agreed as a team to continue with Smart Cart project
- Discussed the main problem and who the system is for

29/3/2026

- Finalized the project idea
- Identified main areas like navigation, sensors, and simulation

31/3/2026

- Roles were assigned in the team
- Started working on the report structure using Overleaf

2/4/2026 – 4/4/2026

- Did research on similar systems and indoor robotics
- Worked on writing the introduction section

6/4/2026 – 8/4/2026

- Spoke with team members to better understand the system components
- Started writing sections related to system overview and environment

9/4/2026 – 12/4/2026

- Worked on documenting system architecture
- Focused on explaining how parts of the system connect together

13/4/2026 – 16/4/2026

- Worked on feasibility and system requirements
- Added safety and performance aspects into the report

17/4/2026 – 20/4/2026

- Defined testing approach based on simulation
- Added different test scenarios into the report

21/4/2026 – 24/4/2026

- Reviewed and improved the report content
- Fixed structure and made sections clearer

25/4/2026 – 28/4/2026

- Final proofreading and formatting
- Prepared the report for submission

Reflection

Working on the documentation helped me understand the system better even though I was not directly involved in implementation. I improved my ability to organize technical information and present it clearly.